

Two-Stage Diffusion Model for 3D Medical Image Segmentation

Masaki Nishimura

*College of Information Science and Engineering
Ritsumeikan University
Osaka, Japan
is0572xs@ed.ritsumei.ac.jp*

Takaya Ueda

*College of Information Science and Engineering
Ritsumeikan University
Osaka, Japan
is0203xf@ed.ritsumei.ac.jp*

Eisuke Ito

*College of Information Science and Engineering
Ritsumeikan University
Osaka, Japan
is0448fs@ed.ritsumei.ac.jp*

Ikuko Nishikawa

*College of Information Science and Engineering
Ritsumeikan University
Osaka, Japan
nishi@ci.ritsumei.ac.jp*

Abstract—Several generative models have been proposed for image segmentation. The latest diffusion models exhibit generation abilities and segmentation superior to those of previous models based on variational autoencoders or generative adversarial nets. However, diffusion models generally require large structures for the reverse diffusion process as well as considerable memory. For the segmentation of three-dimensional (3D) medical volume data, the original data are typically decomposed into multiple two-dimensional (2D) slices or smaller patches as the input for the segmentation model to reduce computational costs. However, this method can limit the receptive field to capture the structure of the original volume data. We propose a novel two-stage diffusion-based segmentation model that retains its original structure with reduced computational costs. In the first stage, coarse detection and segmentation are performed on the entire volume. In the second stage, finer segmentation is performed on each small patch. Furthermore, the prediction map obtained in the first stage is used as a given condition during the denoising process, thus global information is considered for fine segmentation. The proposed method is applied for brain tumor segmentation on the BraTS 2021 dataset and achieved an improved accuracy for multiple types of tumor regions. When focusing on the segmentation of smaller enhancing tumors (ET), the proposed method achieved Dice score of 88.0%, which is 1.6% higher than that of the baseline model, Diff-UNet. This performance signifies a substantial improvement over previous model, especially in the challenging task of brain tumor segmentation.

Index Terms—Generative model, Diffusion probabilistic model, Segmentation, 3D medical data, Brain tumor

I. INTRODUCTION

Segmentation tasks for locating the regions or boundaries of a target object in an image have attracted considerable research attention. Several methods, such as SegNet by Badrinarayanan et al. [1] and SegFormer by Xie et al. [2], have been proposed. However, the segmentation of specific organs or tumors is fairly difficult compared with the segmentation of nonmedical images. Each organ or tumor has a specific and complex shape and structure, and the detailed features of the image may differ depending on the hospital, instrument, and other factors.

Therefore, attaining a segmentation performance comparable to that of the medical experts is difficult.

Several deep generative models have been proposed for image segmentation. Among these models, diffusion probabilistic models (DPMs) [3] exhibit higher generation abilities than those of previous generative models, including variational autoencoders (VAE) [4] and generative adversarial nets (GAN) [5]. DPMs also generate data based on probability distribution that is acquired from a given dataset and can be extended to various applications. For example, DALL-E2 [6] and Stable Diffusion [7] can generate high-quality images with various expressions by specifying the style or topic of the image. Moreover, DPMs have been reported to be effective for various tasks with training labels such as super-resolution [8], anomaly detection [9], and segmentation [10]–[14].

Among these tasks, segmentation is important for medical diagnosis using computer tomography (CT) and magnetic resonance (MR) images. Diffusion models have proven effective for medical image segmentation for various organs, such as liver CT [15] and abdominal CT [16]. The use of expressive diffusion models ensures high segmentation accuracy for complex shapes or small regions. However, the network of the diffusion model generally becomes large, which incurs considerable computational costs for 3D medical data. Therefore, 2D axial slices or sliding local 3D patches are often used as inputs to the model. However, faulty inputs can result in a loss of structural information of an organ.

In this study, stepwise segmentation using a diffusion model is proposed. In the first stage, coarse segmentation considering global features is performed on the entire structure. In the second stage, fine and local segmentation is performed on the coarse segmentation obtained in the previous stage. Low-resolution whole 3D volume data are input into the first stage model to obtain a low-resolution whole segmentation map. Subsequently, a local patch of the obtained low-resolution map is added to the second stage model during the denoising pro-

cess to output a high-resolution local segmentation, which is combined to obtain a final high-resolution whole segmentation. Moreover, the stepwise approach enables the use of a different loss function at each stage to reduce false negatives for a small positive regions in the first stage.

The remainder of this paper is organized as follows: First, Section II briefly introduces related studies and existing methods. Section III explains the proposed stepwise segmentation method using diffusion models. Section IV presents computer experiments on an open dataset of brain tumor MR images, BraTS 2021 dataset [17]–[21] to verify the effectiveness of the proposed method. Finally, Section V presents the conclusion of the paper by summarizing the outcomes.

II. RELATED WORKS

A. Diffusion Probabilistic Models

The diffusion probabilistic model is a generative model based on a Markov chain and is composed of a forward process of diffusion by adding noise and a backward process of reverse diffusion by denoising to generate data. Sohl-Dickstein et al. [22] proposed the first diffusion model that mapped destroyed data on a noise distribution. Then, Ho et al. [3] revealed that the model is equivalent to the score-matching model and introduced the Denoising Diffusion Probabilistic Model (DDPM) through log-likelihood estimation. Several improvements have been proposed in learning and sampling algorithms, including the Denoising Diffusion Implicit Model (DDIM) [23] and distributed scheduling [24].

During the forward diffusion process q , Gaussian noise is repeatedly added to the original image $\mathbf{x}_0$ at t -th step ($t = 1, \dots, T$). The variance β_t of Gaussian noise is increased according to t . Thus, a sequence of image $\mathbf{x}_t$ is generated recursively as follows:

$$q(\mathbf{x}_t | \mathbf{x}_{t-1}) := \mathcal{N}(\mathbf{x}_t; \sqrt{1 - \beta_t} \mathbf{x}_{t-1}, \beta_t \mathbf{I}). \quad (1)$$

Equivalently, an image $\mathbf{x}_t$ at any step t can be sampled from a single Gaussian using

$$q(\mathbf{x}_t | \mathbf{x}_0) = \mathcal{N}(\mathbf{x}_t; \sqrt{\bar{\alpha}_t} \mathbf{x}_0, (1 - \bar{\alpha}_t) \mathbf{I}), \quad (2)$$

where $\alpha_t \equiv 1 - \beta_t$ and $\bar{\alpha}_t \equiv \prod_{s=1}^t \alpha_s$. Moreover, $\mathbf{x}_t$ can be directly obtained by the following equation from $\mathbf{x}_0$ using the reparameterization trick [4]:

$$\mathbf{x}_t = \sqrt{\bar{\alpha}_t} \mathbf{x}_0 + \sqrt{1 - \bar{\alpha}_t} \epsilon, \quad \epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I}). \quad (3)$$

By contrast, the reverse diffusion process p_θ is described as follows:

$$p_\theta(\mathbf{x}_{t-1} | \mathbf{x}_t) = \mathcal{N}(\mathbf{x}_{t-1}; \mu_\theta(\mathbf{x}_t, t), \sigma_\theta^2 \mathbf{I}), \quad (4)$$

where θ is obtained through training. Thus, the denoising network ϵ_θ is trained to estimate the noise ϵ added to $\mathbf{x}_t$ to obtain $\mathbf{x}_{t-1} \sim p_\theta(\mathbf{x}_{t-1} | \mathbf{x}_t)$:

$$\mathbf{x}_{t-1} = \frac{1}{\sqrt{\alpha_t}} \left(\mathbf{x}_t - \frac{1 - \alpha_t}{\sqrt{1 - \bar{\alpha}_t}} \epsilon_\theta(\mathbf{x}_t, t) \right) + \sigma_t \zeta, \quad \zeta \sim \mathcal{N}(\mathbf{0}, \mathbf{I}). \quad (5)$$

In this way, the original $\mathbf{x}_0$ is estimated iteratively as follows, starting with a random sample $\mathbf{x}_T \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$.

Conditional diffusion models enable control of image generation. One approach for guiding a diffusion model is Classifier Guidance [25], which utilizes a pre-trained classifier by adding the gradient of the classifier output to the original score in the denoising process. Another approach is Classifier-Free Guidance [26], which adds class information to the model input. In particular, for medical image generation, a diffusion model using the biological information of a patient was proposed [27].

The direct input of high-dimensional data to diffusion models requires huge computational loads and memory consumption. Latent Diffusion Models [7] introduced VAE to reduce the dimensions. Ho et al. [28] proposed cascaded diffusion models, which were multiple-step methods based on the UNet architecture. Similar to the UNet cascade structure proposed by Isensee et al. [29], the model composed of a cascade pipeline of multiple models that are independently trained in different resolutions. The model generates low-resolution data, which are passed to a super-resolution model to generate high-resolution data by upsampling.

B. Generative Models for Medical Image Segmentation

Several deep neural network models, such as UNet [30] and SegNet [1], with convolutional layered networks have been designed for segmentation. However, the convolution is focused on a local structure with a limited receptive field. Thus, generative models are incorporated into segmentation, and Luc et al. [31] proposed a model that introduced GAN into CNN-based segmentation.

Focusing on medical image segmentation, Li et al. [32] proposed a GAN-based model for brain tumor segmentation, and Xue et al. [33] proposed SegAN, which is an end-to-end adversarial network. SegAN used UNet for segmentation and a multiscale L_1 loss function to acquire global and local positional information between pixels.

SegDiff, proposed by Amit et al. [10], is the first diffusion model proposed for image segmentation. In the forward diffusion process, Gaussian noise is added to the correct segmentation label $\mathbf{x}_0$ at each diffusion step t . During the reverse diffusion process, as displayed in Fig. 1, the segmentation map $\mathbf{x}_t$ is the input to a denoising network to output $\mathbf{x}_{t-1}$ conditioned by the simultaneous input of the original image I . Starting from $\mathbf{x}_T$ sampled from Gaussian, this denoising step is repeated by descending t to 0, and the final output $\hat{\mathbf{x}}_0$ is obtained as an estimated segmentation label. The denoising network ϵ_θ based on UNet is expressed as follows:

$$\epsilon_\theta(\mathbf{x}_t, I, t) = D(E(F(\mathbf{x}_t) + G(I), t), t), \quad (6)$$

where E and D are the encoder and decoder parts in UNet, respectively, with embedding t . Here, F and G are the feature encoders for $\mathbf{x}_t$ and I , respectively. As the final output suffers fair variance because of the Gaussian noise estimation, the average over multiple trials is used for the improvement.

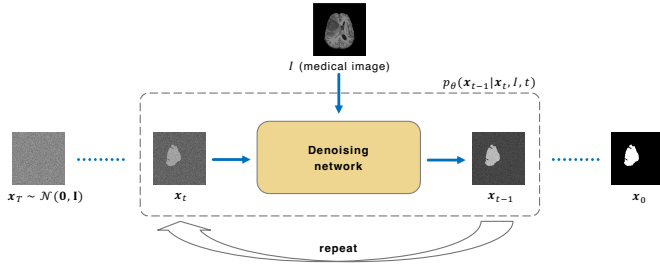


Fig. 1. Reverse diffusion process in segmentation by diffusion models. Starting from random sample x_T from Gaussian, denoising steps are repeated to generate an estimated segmentation label $\hat{x}_0$.

However, this approach is not suitable for complex medical image segmentation.

C. Diff-UNet for 3D Medical Image Segmentation

Wu et al. [34] proposed dynamic conditional encoding as a mutual conditioning between the segmentation map x_t at step t and original image I to improve the segmentation result. Xing et al. [35] extended this approach to 3D medical data and multiple labels to propose Diff-UNet. The denoising network in Diff-UNet has a component called Denoising Module. Fig. 2 displays the architecture of this module. Here, the segmentation map x_t contains multiple channels corresponding to multiple labels, and x_t and I are concatenated as the input to Denoising UNet U . Furthermore, Feature Encoder E^f adds the features of I to each corresponding layer of the encoder in U .

The estimated segmentation label $\hat{x}_0$ obtained by Denoising Module at step t in the reverse diffusion process is expressed as follows:

$$\hat{x}_0 = U(\text{cat}(x_t, I), t, I_f), \quad I_f \sim E^f(I), \quad (7)$$

$$\epsilon_\theta = \frac{x_t - \sqrt{\alpha_t} \hat{x}_0}{\sqrt{1 - \alpha_t}}, \quad (8)$$

where I_f denotes the latent feature values in each layer of the Feature Encoder E^f for the original image I . Thus, Diff-UNet directly outputs the estimation $\hat{x}_0$ through Denoising Module to obtain the denoising network ϵ_θ by (8), which corresponds to (3).

Then, the training of Denoising Module, that is, the training of Denoising UNet U and Feature Encoder E^f , is performed by minimizing the difference between the output $\hat{x}_0$ and the teacher label x_0 measured by the loss function given by the summation of Dice loss, binary cross entropy (BCE) loss, and mean square error (MSE) loss, as follows:

$$\min_{U, E^f} \mathcal{L}_{\text{total}} = \mathcal{L}_{\text{dice}}(\hat{x}_0, x_0) + \mathcal{L}_{\text{bce}}(\hat{x}_0, x_0) + \mathcal{L}_{\text{mse}}(\hat{x}_0, x_0). \quad (9)$$

The denoising process can be accelerated by skipping the reverse diffusion step to decrease the number of repetitions, as is introduced in DDIM. The accuracy is compensated for by the ensemble average over $\hat{x}_0$ obtained by the multiple trials of reverse diffusion. Moreover, the weighted average proposed

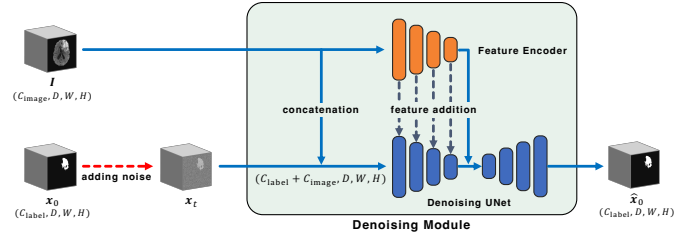


Fig. 2. Architecture of Denoising Module in [35]. The noisy segmentation map x_t and original medical image I are concatenated to be input to Denoising UNet at the t -th step in the reverse diffusion process.

as Step-Uncertainty based Fusion (SUF) [35] improves the accuracy of the final prediction $\bar{x}_0 \equiv$ weighted average of ensemble $\{\hat{x}_0\}$.

We consider that it is difficult to use the whole 3D data as I because of the large structure of Denoising Module. The original study [35] divided original data I into smaller patches with sliding windows for segmentation. Another approach to reduce the data size is segmentation in the latent space of the VAE with lower dimensions [36]. However, it could suffer from overfitting of VAE, especially for the complex shape of the target regions in medical data. This leads to our proposal of stepwise segmentation from coarse to fine and from global to local, where the latter stage is conditioned by the former stage, as in Cascaded Diffusion Models. The next section describes the proposed method.

III. PROPOSED METHOD

A. Overall Framework

The proposed two-stage conditional diffusion model for the segmentation of medical images is described in this section.

In the first stage, a low-resolution whole segmentation map is generated, whereas in the second stage, a high-resolution local segmentation map is generated by conditioning the low-resolution map passed from the first stage. The models are trained independently at each stage. The following two subsections explain the training at each stage more precisely. Although the idea of first coarse and then fine is not novel, the present diffusion model can utilize the first information as the condition in the second stage.

Furthermore, to suppress detection failures or false negatives in the first stage, Tversky loss [37] with an adjustable weight can be used in the training, while the second stage is dedicated to more accurate detection of the boundary.

An overview of the model is presented in Fig. 3. The overall architecture of the diffusion models is based on [24]. Reverse diffusion processes in both stages are implemented by Denoising Module in Diff-UNet, as described in the previous section, which enables multilabel segmentation.

B. First Stage

Fig. 3(a) displays the process in the first stage, where both the original image I and segmentation label x_t are downsampled to be low-resolution data keeping a entire structure.

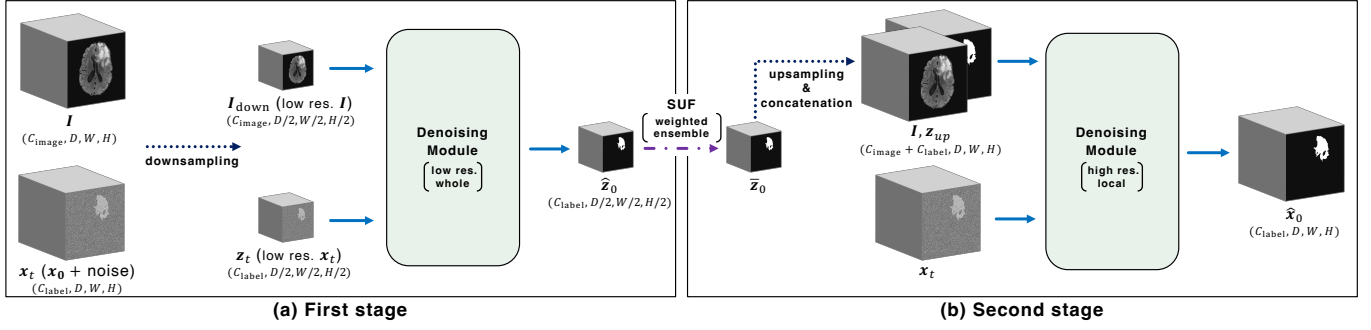


Fig. 3. Overview of the proposed model. The first stage generates a low-resolution global segmentation map, while the second stage generates a high-resolution local segmentation map conditioned by the low-resolution map obtained in the first stage.

The original image $I \in \mathbb{R}^{C_{\text{image}} \times D \times W \times H}$ are downsampled to $I_{\text{down}} \in \mathbb{R}^{C_{\text{image}} \times \frac{D}{2} \times \frac{W}{2} \times \frac{H}{2}}$, and the same is used for the segmentation label x_t . Subsequently, the low-resolution segmentation map $\hat{z}_0 \in \mathbb{R}^{C_{\text{label}} \times \frac{D}{2} \times \frac{W}{2} \times \frac{H}{2}}$ is obtained using the Denoising Module as follows:

$$\hat{z}_0 = U_1(\text{cat}(z_t, I_{\text{down}}), t, I_f), \quad I_f \sim E_1^f(I_{\text{down}}). \quad (10)$$

The use of downscaled image as network inputs resulted in a reduction of memory consumption to 1/8.

Training for Denoising UNet U_1 and Feature Encoder E_1^f in the first stage is performed by minimizing the following loss function with Tversky loss [37] as the first term, instead of the ordinary symmetric Dice loss used in (9):

$$\min_{U_1, E_1^f} \mathcal{L}_{\text{total}} = \mathcal{L}_{\text{tversky}}(\hat{z}_0, z_0) + \mathcal{L}_{\text{bce}}(\hat{z}_0, z_0) + \mathcal{L}_{\text{mse}}(\hat{z}_0, z_0). \quad (11)$$

The first term $\mathcal{L}_{\text{tversky}}$ is defined as follows:

$$\mathcal{L}_{\text{tversky}} = 1 - \frac{TP}{TP + a \cdot FP + b \cdot FN}, \quad (12)$$

where TP , FP and FN indicate the numbers of true positive, false positive and false negative voxels, respectively. When $a = b = 0.5$, this corresponds to the ordinary Dice loss.

After training, a low-resolution segmentation map $\bar{z}_0 \equiv \text{weighted average of the ensemble } \{\hat{z}_0\}$ is obtained and passed to the second stage for both the training and estimation phases.

C. Second Stage

Fig. 3(b) displays the process in the second stage, where both image I and segmentation map x_t maintain the original resolution for the precise segmentation, conditioned by the coarse and whole segmentation map $\bar{z}_0$ passed from the first stage. Here, $\bar{z}_0$ is upsampled to z_{up} . Denoising Module operates on each local patch with sliding windows as follows and is combined to reconstruct the entire structure:

$$\hat{x}_0 = U_2(\text{cat}(x_t, I, z_{\text{up}}), t, I_f), \quad I_f \sim E_2^f(\text{cat}(I, z_{\text{up}})). \quad (13)$$

Although the receptive field of Denoising Module is limited, the global information of each patch is conveyed, not only from the condition z_{up} but also from latent feature I_f .

The training of Denoising UNet U_2 and Feature Encoder E_2^f in the second stage is performed by minimizing the ordinary loss function (9) using Dice loss. The final segmentation $\bar{x}_0$ is obtained by the weighted average over multiple trials.

IV. EXPERIMENT

A. Dataset

The proposed method is applied to the BraTS 2021 training dataset [17]–[21], which is a dataset prepared for Brain Tumor Segmentation (BraTS) Challenge 2023. The dataset contains brain MR imaging (MRI) scans of 1251 subjects with the segmentation of brain tumor sub-regions with labeled data provided by experts. The sub-regions are enhancing tumor (ET), tumor core (TC), and whole tumor (WT). MRI scans have four modalities namely native (T1), post-contrast T1-weighted (T1Gd), T2-weighted (T2), and T2 fluid attenuated inversion recovery (T2-FLAIR). Dice score and 95% Hausdorff distance are used for the evaluation in accordance with the official evaluation measures in the challenge using the official program. The data are preprocessed by co-registration, interpolated to the same resolution, and skull-stripped with a volume size of $155 \times 240 \times 240$.

To compare the proposed method with DBTrans [38], which is a state-of-the-art transformer-based segmentation model in which both local and global features are used, the same dataset is divided into three groups for training, evaluation, and testing in the same way as [38].

B. Implementation Details

Denoising Module, which includes Denoising UNet and Feature Encoder, has identical architecture at both stages. Each network is based on the framework in [46]. Both the encoder and decoder of Denoising UNet, and Feature Encoder are five layers. Furthermore, each layer of Denoising UNet receives feature I_f from the corresponding layer in Feature Encoder and information regarding step t .

In ordinary data preprocessing, resizing, rotation in the first stage, and flipping in the second stage are randomly applied.

Downsampling in the first stage and upsampling in the second stage are performed using tri-linear interpolation. The patch size in the second stage is $96 \times 96 \times 96$. Each patch for

TABLE I

COMPARISON OF THE SEGMENTATION RESULTS BY VARIOUS METHODS MEASURED BY DICE SCORE AND 95% HAUSDORFF DISTANCE FOR THREE SUB-REGIONS AND THE AVERAGE. THE BEST VALUE FOR EACH IS SHOWN IN BOLD. THE RESULTS ABOVE THE HORIZONTAL LINE ARE QUOTED FROM [38].

Method	Dice Score				95% Hausdorff Distance			
	ET	TC	WT	AVG	ET	TC	WT	AVG
3D U-Net [39]	83.39	86.28	89.59	86.42	6.15	6.18	11.49	7.94
Swin-Unet [40]	83.34	87.62	89.81	89.61	6.19	6.35	11.53	8.03
TransBTS [41]	80.35	85.35	89.25	84.99	7.83	8.21	15.12	10.41
UNETR [42]	79.78	83.66	90.10	84.51	9.72	10.01	15.99	11.90
nnFormer [43]	82.83	86.48	90.37	86.56	8.00	7.89	11.66	9.18
VT-Unet-B [44]	85.59	87.41	91.02	88.07	6.23	6.29	10.03	7.52
NestedFormer [45]	85.62	88.18	90.12	87.88	6.08	6.43	10.23	7.63
DBTrans [38]	86.70	90.26	92.41	89.69	6.13	6.24	9.84	7.38
Diff-UNet [35]	86.40	92.13	92.51	90.35	15.63	5.53	7.51	9.55
Ours method	88.02	91.57	92.90	90.83	8.67	7.30	5.12	7.03

the training is randomly cropped, and sliding windows with an overlapping ratio 0.5 are used for the evaluation.

The various parameter values are identical for both stages. The hyperparameters in DDPM is set according to [25], and the sampling step in DDIM is set to 50. The ensemble size is 5.

For the network training, Adams optimization is used with a weight decay of 10^{-5} [47]. The learning rate is controlled by linear warmup and Cosine Annealing [48]. The initial learning rate is set to 10^{-4} . The warmup period is 30 epochs. The batch size is 12. The training period is 500 epochs, with evaluation every 50 epochs. NVIDIA RTX A6000 GPUs are used for the experiments.

The weights of Tversky loss used for the training in the first stage are set $a = 0.3$ and $b = 0.7$ for false positives and false negatives, respectively. The effects of the weights are discussed in the ablation study in IV-D.

C. Results

Table I presents a comparison of the experimental results obtained by Zeng et al. [38] with our results of Diff-UNet and the proposed method. Dice score measures the overlapping ratio between the ground truth area and the segmented area. A larger value is better, with the maximum value of 100(%) attained by the perfect match. 95% Hausdorff distance measures the distance between the above two areas. Smaller values are better, with the minimum value of zero attained by the perfect match. The results above the horizontal line in Table I is quoted from [38], where descending/ascending orders of 95% Hausdorff distance for ET, TC, and WT sub-regions are reverse to those in this study. This may indicate that the algorithm of the measurement is different. Our program is obtained from the official website of the Challenge, and the resulting orders are consistent with official documents [49].

The results in Table I reveals that the proposed method attains the highest Dice scores for ET and WT sub-regions, and thus for the average and also the highest Hausdorff distance for the WT and the average. Especially for ET, which is the smallest sub-region, the proposed method improves Dice score by 1.62 compared with Diff-UNet, on which the proposed method is based. This result indicates the effectiveness of dividing the roles into two stages, that is, in the first stage,

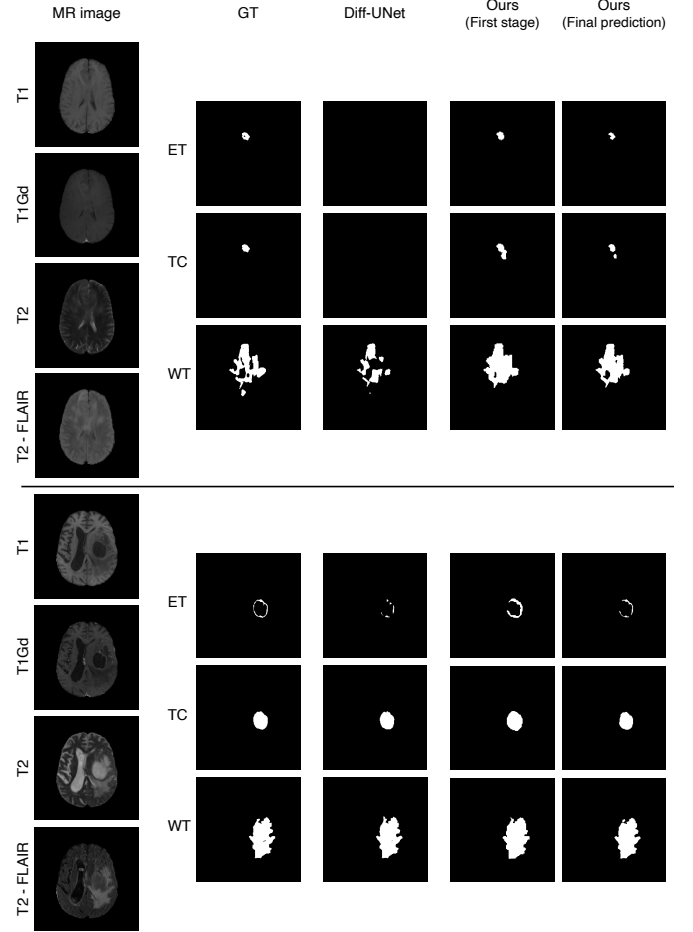


Fig. 4. Comparison of the segmentation maps by Diff-UNet and the proposed method for ET, TC, and WT sub-regions in two examples of slices. Left end images are MRI scans of four modalities.

the tumor is detected in the whole brain with suppressed false negatives, whereas in the second stage, the tumor is segmented precisely under the condition.

Fig. 4 presents a comparison of the segmentation maps for two examples of MRI slices obtained using Diff-UNet and the proposed method. For the upper example slice with small ET and TC regions, the proposed method segments them,

TABLE II
COMPARISON OF SEGMENTATION RESULTS DEPENDING ON THE WEIGHTS IN TVERSKY LOSS IN THE FIRST STAGE

Weight Coefficients a and b for Tversky Loss in the first stage	ET			TC			WT			AVG		
	Dice	Sn	Sp	Dice	Sn	Sp	Dice	Sn	Sp	Dice	Sn	Sp
$a = b = 0.5$ (Dice loss)	87.17	87.42	99.98	91.68	91.76	99.98	92.54	92.44	99.95	90.46	90.54	99.97
$a = 0.4, b = 0.6$	87.38	87.51	99.98	91.99	91.84	99.98	92.77	92.80	99.95	90.72	90.71	99.97
$a = 0.3, b = 0.7$ (best)	88.02	87.93	99.98	91.57	91.74	99.98	92.90	92.64	99.95	90.83	90.77	99.97
$a = 0.2, b = 0.8$	87.93	88.71	99.98	91.74	91.91	99.98	92.37	92.90	99.94	90.68	91.17	99.97

larger in the first stage and more accurately in the second stage, whereas Diff-UNet does not. For the lower example, with a more complex ET region shape, the proposed method segments more precisely than Diff-UNet.

However, the results for TC in Table I are inferior than those of Diff-UNet. Fig. 4 displays our false detections. The reason for our worse results is that, originally, there are fewer false negatives for TC, whereas Tversky loss in the first stage increases false detections.

D. Ablation Study

The effect of Tversky loss used in the first study is examined by comparing the results with various weight coefficient settings. The weights a and b in (12) for the false positives and false negatives, respectively, are varied in the first stage, and the segmentation maps obtained in the second stage are evaluated by Dice score, sensitivity (Sn), and specificity (Sp). A larger b suppresses false negatives; therefore, in general, higher sensitivity and lower specificity are expected. What should be noted here is the effect of the weights in the first stage on the final results in the second stage.

The results are summarized in Table II. Tversky loss is found to effectively improve the final results. The third row is the same as the result in Table I, which is the best weight, as reported in [37]. However, a smaller b seems appropriate for the TC sub-region, which is consistent with the discussion in IV-C. Therefore, the different settings for each sub-region is expected to improve the total accuracy.

V. CONCLUSION

In this study, a novel two-stage diffusion model is proposed for 3D medical image segmentation. The coarse and whole segmentation map generated in the first stage is used as the conditions in the second stage, where fine and local segmentation maps are generated. Thus, the precise segmentation of each local patch with a lower computational cost is attained by the conditional diffusion process. The effectiveness is verified through computer experiments on an open dataset of 3D brain tumors, the BraTS 2021 dataset. The proposed method achieves a higher performance, especially for a small sub-region of the tumor.

Further improvements to the model include end-to-end learning optimization. In addition, the model will be applied to other types of medical images and other organs.

REFERENCES

- [1] V. Badrinarayanan, A. Kendall, and R. Cipolla, "Segnet: a deep convolutional encoder-decoder architecture for image segmentation," *IEEE transactions on pattern analysis and machine intelligence*, vol. 39, no. 12, pp. 2481–2495, 2017.
- [2] E. Xie, W. Wang, Z. Yu, A. Anandkumar, J. M. Alvarez, and P. Luo, "SegFormer: simple and efficient design for semantic segmentation with transformers," *Advances in Neural Information Processing Systems*, vol. 34, pp. 12 077–12 090, 2021.
- [3] J. Ho, A. Jain, and P. Abbeel, "Denoising diffusion probabilistic models," *Advances in neural information processing systems*, vol. 33, pp. 6840–6851, 2020.
- [4] D. P. Kingma and M. Welling, "Auto-encoding variational bayes," in *2nd International Conference on Learning Representations, ICLR 2014, Banff, AB, Canada, April 14-16, 2014, Conference Track Proceedings*, 2014.
- [5] I. Goodfellow, J. Pouget-Abadie, M. Mirza, B. Xu, D. Warde-Farley, S. Ozair, A. Courville, and Y. Bengio, "Generative adversarial nets," *Advances in neural information processing systems*, vol. 27, 2014.
- [6] A. Ramesh, P. Dhariwal, A. Nichol, C. Chu, and M. Chen, "Hierarchical text-conditional image generation with clip latents," *arXiv preprint arXiv:2204.06125*, 2022.
- [7] R. Rombach, A. Blattmann, D. Lorenz, P. Esser, and B. Ommer, "High-resolution image synthesis with latent diffusion models," in *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, 2022, pp. 10 684–10 695.
- [8] C. Saharia, J. Ho, W. Chan, T. Salimans, D. J. Fleet, and M. Norouzi, "Image super-resolution via iterative refinement," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 45, no. 4, pp. 4713–4726, 2022.
- [9] J. Wolleb, F. Bieder, R. Sandkühler, and P. C. Cattin, "Diffusion models for medical anomaly detection," in *International Conference on Medical image computing and computer-assisted intervention*. Springer, 2022, pp. 35–45.
- [10] T. Amit, E. Nachmani, T. Shaharabany, and L. Wolf, "Segdiff: image segmentation with diffusion probabilistic models," *arXiv preprint arXiv:2112.00390*, 2021.
- [11] D. Baranchuk, A. Voynov, I. Rubachev, V. Khurlov, and A. Babenko, "Label-efficient semantic segmentation with diffusion models," in *International Conference on Learning Representations, ICLR*, 2022.
- [12] J. Xu, S. Liu, A. Vahdat, W. Byeon, X. Wang, and S. De Mello, "Open-vocabulary panoptic segmentation with text-to-image diffusion models," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2023, pp. 2955–2966.
- [13] Y. Ji, Z. Chen, E. Xie, L. Hong, X. Liu, Z. Liu, T. Lu, Z. Li, and P. Luo, "Ddp: diffusion model for dense visual prediction," in *2023 IEEE/CVF International Conference on Computer Vision, ICCV*, 2023, pp. 21 684–21 695.
- [14] W. Tan, S. Chen, and B. Yan, "Diffss: diffusion model for few-shot semantic segmentation," *arXiv preprint arXiv:2307.00773*, 2023.
- [15] A. Rahman, J. M. J. Valanarasu, I. Hacıhaliloglu, and V. M. Patel, "Ambiguous medical image segmentation using diffusion models," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2023, pp. 11 536–11 546.
- [16] J. Wu, R. Fu, H. Fang, Y. Zhang, and Y. Xu, "Medsegdiff-v2: diffusion based medical image segmentation with transformer," *arXiv preprint arXiv:2301.11798*, 2023.
- [17] U. Baid, S. Ghodasara, S. Mohan, M. Bilello, E. Calabrese, E. Colak, K. Farahani, J. Kalpathy-Cramer, F. C. Kitamura, S. Pati *et al.*, "The RSNA-ASNR-MICCAI BraTS 2021 benchmark on brain tumor segmentation and radiogenomic classification," *arXiv preprint arXiv:2107.02314*, 2021.

- [18] B. H. Menze, A. Jakab, S. Bauer, J. Kalpathy-Cramer, K. Farahani, J. Kirby, Y. Burren, N. Porz, J. Slotboom, R. Wiest *et al.*, “The multimodal brain tumor image segmentation benchmark (BRATS),” *IEEE transactions on medical imaging*, vol. 34, no. 10, pp. 1993–2024, 2014.
- [19] S. Bakas, H. Akbari, A. Sotiras, M. Bilello, M. Rozycki, J. S. Kirby, J. B. Freymann, K. Farahani, and C. Davatzikos, “Advancing the cancer genome atlas glioma MRI collections with expert segmentation labels and radiomic features,” *Scientific data*, vol. 4, no. 1, pp. 1–13, 2017.
- [20] S. Bakas, H. Akbari, A. Sotiras, M. Bilello, M. Rozycki, J. Kirby, J. Freymann, K. Farahani, and C. Davatzikos, “Segmentation labels and radiomic features for the pre-operative scans of the TCGA-GBM collection,” *The cancer imaging archive*, 2017.
- [21] S. Bakas, H. Akbari, A. Sotiras, M. Bilello, M. Rozycki, J. Kirby, J. Freymann, K. Farahani, and C. Davatzikos, “Segmentation labels and radiomic features for the pre-operative scans of the TCGA-LGG collection,” *The cancer imaging archive*, 2017.
- [22] J. Sohl-Dickstein, E. Weiss, N. Maheswaranathan, and S. Ganguli, “Deep unsupervised learning using nonequilibrium thermodynamics,” in *International conference on machine learning*. PMLR, 2015, pp. 2256–2265.
- [23] J. Song, C. Meng, and S. Ermon, “Denoising diffusion implicit models,” in *International Conference on Learning Representations*, 2021.
- [24] A. Q. Nichol and P. Dhariwal, “Improved denoising diffusion probabilistic models,” in *International Conference on Machine Learning*. PMLR, 2021, pp. 8162–8171.
- [25] P. Dhariwal and A. Nichol, “Diffusion models beat gans on image synthesis,” *Advances in neural information processing systems*, vol. 34, pp. 8780–8794, 2021.
- [26] J. Ho and T. Salimans, “Classifier-free diffusion guidance,” in *NeurIPS 2021 Workshop on Deep Generative Models and Downstream Applications*, 2021.
- [27] W. H. Pinaya, P.-D. Tudosiu, J. Dafflon, P. F. Da Costa, V. Fernandez, P. Nachev, S. Ourselin, and M. J. Cardoso, “Brain imaging generation with latent diffusion models,” in *MICCAI Workshop on Deep Generative Models*. Springer, 2022, pp. 117–126.
- [28] J. Ho, C. Saharia, W. Chan, D. J. Fleet, M. Norouzi, and T. Salimans, “Cascaded diffusion models for high fidelity image generation,” *The Journal of Machine Learning Research*, vol. 23, no. 1, pp. 2249–2281, 2022.
- [29] F. Isensee, P. F. Jaeger, S. A. Kohl, J. Petersen, and K. H. Maier-Hein, “nnU-Net: a self-configuring method for deep learning-based biomedical image segmentation,” *Nature methods*, vol. 18, no. 2, pp. 203–211, 2021.
- [30] O. Ronneberger, P. Fischer, and T. Brox, “U-net: convolutional networks for biomedical image segmentation,” in *Medical Image Computing and Computer-Assisted Intervention–MICCAI 2015: 18th International Conference, Munich, Germany, October 5–9, 2015, Proceedings, Part III 18*. Springer, 2015, pp. 234–241.
- [31] P. Luc, C. Couprie, S. Chintala, and J. Verbeek, “Semantic segmentation using adversarial networks,” in *NeurIPS 2016 Adversarial Training Workshop*, 2016.
- [32] Z. Li, Y. Wang, and J. Yu, “Brain tumor segmentation using an adversarial network,” in *Brainlesion: Glioma, Multiple Sclerosis, Stroke and Traumatic Brain Injuries: Third International Workshop, BrainLes 2017, Held in Conjunction with MICCAI 2017, Quebec City, QC, Canada, September 14, 2017, Revised Selected Papers 3*. Springer, 2018, pp. 123–132.
- [33] Y. Xue, T. Xu, H. Zhang, L. R. Long, and X. Huang, “SegAN: adversarial network with multi-scale L_1 loss for medical image segmentation,” *Neuroinformatics*, vol. 16, pp. 383–392, 2018.
- [34] J. Wu, R. FU, H. Fang, Y. Zhang, Y. Yang, H. Xiong, H. Liu, and Y. Xu, “MedSegDiff: medical image segmentation with diffusion probabilistic model,” in *Medical Imaging with Deep Learning*, 2023.
- [35] Z. Xing, L. Wan, H. Fu, G. Yang, and L. Zhu, “Diff-UNet: a diffusion embedded network for volumetric segmentation,” *arXiv preprint arXiv:2303.10326*, 2023.
- [36] H. Vu Quoc, T. Tran Le Phuong, M. Trinh Xuan, and S. Dinh Viet, “LSegDiff: a latent diffusion model for medical image segmentation,” in *Proceedings of the 12th International Symposium on Information and Communication Technology*, 2023, pp. 456–462.
- [37] S. S. M. Salehi, D. Erdogmus, and A. Gholipour, “Tversky loss function for image segmentation using 3D fully convolutional deep networks,” in *International workshop on machine learning in medical imaging*. Springer, 2017, pp. 379–387.
- [38] X. Zeng, P. Zeng, C. Tang, P. Wang, B. Yan, and Y. Wang, “DBTrans: a dual-branch vision transformer for multi-modal brain tumor segmentation,” in *International Conference on Medical Image Computing and Computer-Assisted Intervention*. Springer, 2023, pp. 502–512.
- [39] Ö. Çiçek, A. Abdulkadir, S. S. Lienkamp, T. Brox, and O. Ronneberger, “3D U-Net: learning dense volumetric segmentation from sparse annotation,” in *Medical Image Computing and Computer-Assisted Intervention–MICCAI 2016: 19th International Conference, Athens, Greece, October 17–21, 2016, Proceedings, Part II 19*. Springer, 2016, pp. 424–432.
- [40] H. Cao, Y. Wang, J. Chen, D. Jiang, X. Zhang, Q. Tian, and M. Wang, “Swin-Unet: unet-like pure transformer for medical image segmentation,” in *European conference on computer vision*. Springer, 2022, pp. 205–218.
- [41] W. Wang, C. Chen, M. Ding, H. Yu, S. Zha, and J. Li, “TransBTS: multimodal brain tumor segmentation using transformer,” in *Medical Image Computing and Computer Assisted Intervention–MICCAI 2021: 24th International Conference, Strasbourg, France, September 27–October 1, 2021, Proceedings, Part I 24*. Springer, 2021, pp. 109–119.
- [42] A. Hatamizadeh, Y. Tang, V. Nath, D. Yang, A. Myronenko, B. Landman, H. R. Roth, and D. Xu, “UNETR: transformers for 3D medical image segmentation,” in *Proceedings of the IEEE/CVF winter conference on applications of computer vision*, 2022, pp. 574–584.
- [43] H.-Y. Zhou, J. Guo, Y. Zhang, L. Yu, L. Wang, and Y. Yu, “nnFormer: interleaved transformer for volumetric segmentation,” *arXiv preprint arXiv:2109.03201*, 2021.
- [44] H. Peiris, M. Hayat, Z. Chen, G. Egan, and M. Harandi, “A robust volumetric transformer for accurate 3D tumor segmentation,” in *International Conference on Medical Image Computing and Computer-Assisted Intervention*. Springer, 2022, pp. 162–172.
- [45] Z. Xing, L. Yu, L. Wan, T. Han, and L. Zhu, “NestedFormer: nested modality-aware transformer for brain tumor segmentation,” in *International Conference on Medical Image Computing and Computer-Assisted Intervention*. Springer, 2022, pp. 140–150.
- [46] T. Falk, D. Mai, R. Bensch, Ö. Çiçek, A. Abdulkadir, and Y. Marrakchi, “U-net: deep learning for cell counting, detection, and morphometry,” *Nature methods*, vol. 16, pp. 67–70, 2019.
- [47] I. Loshchilov and F. Hutter, “Decoupled weight decay regularization,” in *International Conference on Learning Representations*, 2019.
- [48] I. Loshchilov and F. Hutter, “SGDR: stochastic gradient descent with warm restarts,” in *International Conference on Learning Representations, ICLR*, 2017.
- [49] R. A. Zeineldin, M. E. Karar, O. Burgert, and F. Mathis-Ullrich, “Multimodal CNN networks for brain tumor segmentation in MRI: a BraTS 2022 challenge solution,” in *Brainlesion: Glioma, Multiple Sclerosis, Stroke and Traumatic Brain Injuries*. Springer Nature Switzerland, 2023, pp. 127–137.